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Full length article

Streamlined robotic hand–eye calibration of multiple 2D-profilers: A rapid, closed-form two-stage method via a single-plane artefact

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ABSTRACT

A 2D laser profiler is commonly utilized in high-precision robotic settings to capture detailed surface profiles for 3D scanning. By collecting and combining numerous such measurements from different viewpoints, it is possible to assemble a comprehensive 3D map. However, to effectively merge these individual 2D profiles into a singular global framework, the spatial relationship between the scanners and the robot's reference frame is required. Traditional hand–eye calibration techniques typically necessitate specific calibration artifacts or extraneous positional sensors, and the process is either manually executed or only partially automated, demanding considerable time and effort. This paper introduces an innovative, closed-form approach to hand–eye calibration that can be applied to a single scanner or an array of multiple scanners. Our method circumvents the requirements for initial parameter estimates or specialized calibration implements, instead employing a flat plane for hand–eye calibration. This method paves the way for a fully automated calibration sequence comprising only three rotational and three translational poses, reducing the total calibration duration. This streamlined process has undergone strict experimental validation utilizing a calibrated sphere, proving its effectiveness not only with a solitary scanner setup but also with an ensemble of three scanners.

1. Introduction

3D machine vision is a powerful tool that generates accurate digital models of complex objects [1,2]. It is applied in inspecting automotive parts [3], verifying packaging [4,5], pick and place [6,7], object tracking, sorting [8], guiding robots [9–11] and additive manufacturing [12]. In manufacturing, these models aid in reverse engineering, design, prototyping, quality control, inspection, toolpath optimization, process management, and digital twin.

Various 3D vision sensors are available in the industry based on the technology used. The principal techniques are time-of-flight (TOF), stereo vision, laser triangulation, and structured light. The distinguishing features for selecting the sensor are the required range and accuracy of the measurement. Laser triangulation 3D vision systems are typically used for inspection and metrology applications. In this category, the single-line laser sensor, also known as a 2D laser profiler, is generally used for scanning moving objects. The 2D laser profiler captures and fuses object-to-sensor distances directly, delivering high-fidelity, high-resolution data crucial for the examination of surface flaws and dimensional accuracy [13,14].

With the growth of Industry 4.0, the increase of automation across different sectors [15] makes laser scanning essential for inline metrology, especially when combined with robotic arms for tasks assisted by robotics [16,17]. A 2D laser profiler on a robotic arm or multiple profilers targeted at moving robot hand enables 3D reconstruction by merging numerous line scans for 3D model creation [18]. Precise mapping of the global coordinate system is necessary for accurate reconstructions [19], for which hand–eye calibration is a prerequisite.

Hand–eye calibration is essential to define the spatial relationship between the base of a robot and the sensor coordinate frame. Specifically, eye-to-hand calibration is applied when the robot arm manipulates the target with stationary sensors. At the same time, eye-in-hand calibration describes a mobile sensor attached to the robotic arm. The hand–eye calibration process for all vision systems mentioned above has 3D information regarding the 2D grid and intensity, color, or depth value. The corresponding image feature is used to perform the hand–eye calibration. In contrast, line scanners face difficulty consistently identifying fixed feature points from 2D range data and, therefore, to be moved and re-scanned in several poses and orientations, substantially increasing the data capture time. Additionally,

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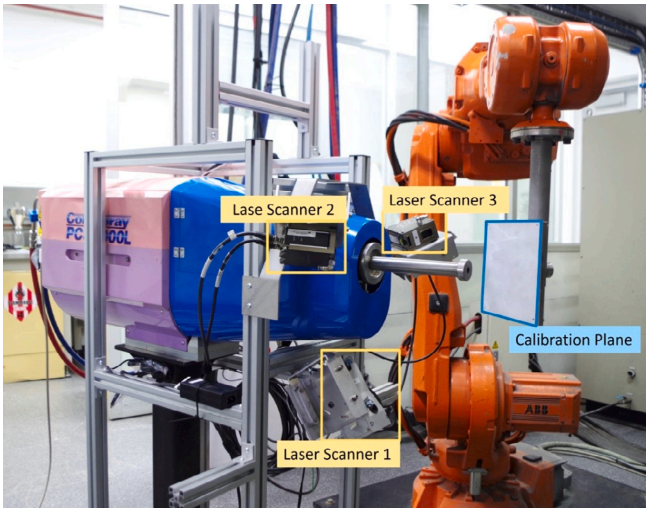


Fig. 1. Experimental setup for real-time scanning with multiple scanners in cold spray additive manufacturing lab.

repeated calibration processes are performed in industrial environments where robots, sensors, workpieces, and tools are frequently reconfigured or re-positioned. These factors increase the setup time and demand specialized knowledge, presenting operational inefficiencies.

This paper introduces an innovative closed-form hand–eye calibration technique for single and multiple scanners. It is the first time a two-step method utilizing a single plane has been formulated and studied for 2D line profilers without relying on edge/feature detection. Significantly, this approach eliminates the need for preliminary estimations or specialized calibration equipment. Benefiting from the advantages of the two-step process, it optimally uses a flat plane to calculate the transformation between the robot base and the sensor's fixed coordinate system. This method provides a fully automated calibration solution, simplifying the process by minimizing six robot poses(three translational and three rotational poses). Experimental validation is conducted for two use cases, i.e., a single scanner and a trio of scanners, employing a calibrated sphere as a reference. The advantages of the method are the following.

- *Accessible calibration artifact*: it utilizes an accessible planar plate, seamless integration, ensuring practicality.
- *Novel formulation*: a novel two-step closed-form formulation establishing rigid body relationship of laser rays intersection with a plane at various locations.
- *No initial estimation requirement*: no prior estimation of the transformation parameter of the scanner is needed.
- *Minimal data collection*: minimizes the required data, streamlining the data collection for efficiency.
- *Rapid calibration*: reducing setup reconfiguration and pose acquisition times.
- *Single or multiple 2D profilers hand–eye calibration*

2. Related work

The challenge of hand–eye calibration has been a persistent area of interest alongside the burgeoning use of robotics in various manufacturing industries since the 1980s [19]. The calibration process for 2D line scanners could be categorized based on the calibration artifact used, the algorithm employed, and the level of user engagement. Table 1 summarizes existing methods. These elements determined the calibration's efficiency, time consumption, and cost-effectiveness.

Although straightforward, simple calibration targets, such as crosshair marks and sharp tips, required expert visual oversight, rendering the process laborious and inefficient [19]. On the contrary, sophisticated bespoke targets, such as disks, stepped surfaces, and spherical objects [20,21], facilitated automation in calibration but were hampered by limited availability, production complexity, and high costs. With two [22] or three [23] distinct orthogonal planes intuitively reduced the required measurements by taking advantage of planar and orthogonal constraints [22–24]. However, such a multi-planar technique proves impractical in eye-to-hand calibrations, where the robot's capacity to manipulate complex artifacts is a prerequisite.

A single planar surface, by comparison, has been ubiquitously accessible within robotic setups, economically viable, and adept at facilitating automated calibration. Hand–eye calibration involving planar artifacts has been studied applying iterative approaches [25–27]. Some robotic tasks involving object manipulation, such as additive manufacturing, require the deployment of multifaceted 2D scanners for complete 3D scanning. This requisite requires efficient and accurate extrinsic calibration of all sensors. Consequently, selecting a calibration artifact and algorithms is an integral decision in optimizing the calibration process according to the specific robotic configuration.

Calibration algorithms can be divided into iterative, closed-form separable solutions (two-step) and closed-form simultaneous solutions (one-step). The iterative approach gradually refines the transformation parameter based on the optimization algorithm and the quality and diversity of collected data [23,25,27]. The closed-form solution [20,30] is preferable to the iterative approach due to its constant computational cost and easy variable result analysis. Although the one-step approach of recovering rotation and translation components is appealing, as the error in one step is not propagated to the next, the result varies with the scaling of the positional component [31]. [32] applied a two-step closed-form method, outperforming optimization-based methods in terms of computational cost and maintaining the quality of the result. [30] used an edge-based specimen with a two-step least squares approach for hand–eye calibration and validated their method by fitting a sphere with satisfactory results. However, We observed from our experiment that line scanners suffered measurement accuracy around the edges, especially when the angle of incidence is non-perpendicular to the edge in reflective surfaces. No two-step calibration method in the literature utilizing a single plane without edge detection exists.

There are two common types of hand–eye calibration based on the mounting position of the scanner, namely eye-in-hand configuration and eye-to-hand configuration. In the eye-in-hand configuration, the sensor is mounted directly on the robot's end effector, which moves with the robot's movement. Hence, the eye-in-hand calibration solves the unknown relationship between the robot base and the workobject. This configuration is common for 3D scanning applications [19,28]. Also, the laser profiler on the robot arm provides versatile in-situ monitoring ability for metal additive manufacturing [13,33,34], and robotic welding [35], where the workobject is stationary and the tool is moving. In contrast, eye-to-hand calibration is performed to find the sensor pose relative to the robot base coordinate frame. This configuration is suitable for tasks where the sensor needs a constant view of the entire workspace, such as process monitoring or inspection. A stationary laser profiler is utilized for inspection in the grinding application of an aero-space blade [20].

3. Theoretical background

The experimental setup for in-process scanning in a cold spray additive manufacturing lab is represented by Fig. 1. The setup comprises an industrial robot equipped with a flat plane substrate dedicated as a base for the 3D printing of objects. The same planar base is also used as an artifact for calibration. Additionally, the system features a stationary tool, i.e., a cold spray gun and multiple stationary scanners oriented towards the substrate.

Fig. 2 presents an illustrative depiction of a robotic additive manufacturing setup. Each constituent element is distinctly labeled with

Table 1
Summary of 2D profiler hand–eye calibration methods based on the artifact shape/accessibility, solution type, and need for an initial estimate.

Method	Year	Artefact	Solution	Initial estimate	User intervention	Number of poses
Yin [19]	2013	cross-hair	Iterative	Yes	High	60
Carlson [26]	2015	1-plane	Iterative	Yes	High	50
Wagner [28]	2015	Tip	Manual	No	Semi	30
Xu [20]	2017	Sphere	Two-step	No	High	32
Liska [29]	2018	Multi-plane	Iterative	Yes	High	1000
Lembono [27]	2019	1-plane	Iterative	Yes	High	150
Sharifzadeh [25]	2020	1-plane	Iterative	Yes	Semi	48
Al Khawli [23]	2021	3-plane	Iterative	Yes	High	38
Paschke [21]	2022	plane/sphere	Iterative	Yes	Semi	240–360
Xu [30]	2022	Edge	Two-step LS	No	Semi	250
Our Method	2023	1-plane	Two-step	No	No	6

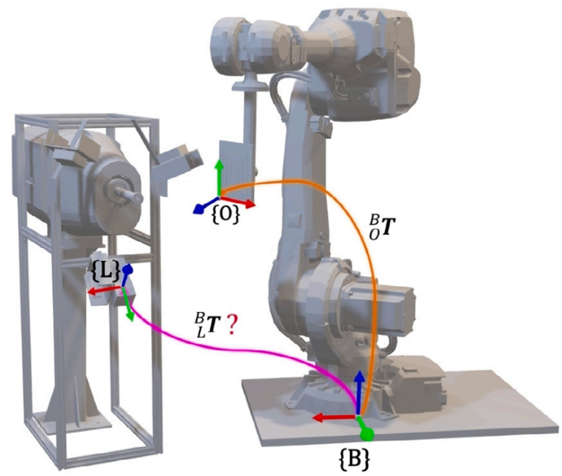


Fig. 2. A robotic eye-to-hand configuration and coordinate frames of laser scanner, robot base, and work object.

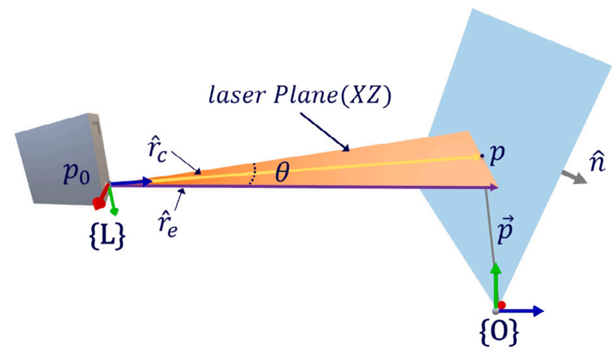


Fig. 3. Projection of laser rays within laser opening angle θ illustrating central laser ray and edge laser ray.

its local coordinate frame: the robot base **B**, the work object **O**, and the laser sensor **L**. The given equation satisfies the transformation that correlates any observed point in a 2D profile to its corresponding point on the calibration plane:

$$^{O}P = {}^{O}_{L}\boldsymbol{T}\,{}^{L}P = {}^{O}_{B}\boldsymbol{T}\,{}^{B}_{L}\boldsymbol{T}\,{}^{L}P. \tag{1}$$

Here, ^{L}P encompasses all points acquired by a scanner, denoted as $^{L}P = {}^{L}_{1}p, {}^{L}_{2}p, \ldots, {}^{L}_{n}p$, and ^{O}P represents the projection of entire laser beam onto the calibration plane, given by $^{O}P = {}^{O}_{1}p, {}^{O}_{2}p, \ldots, {}^{O}_{n}p$. The kinematic chain involves the transformation from the robot base frame to the work object, specifically the substrate plate, denoted as ${}^{O}_{B}\boldsymbol{T}$, determined by the manipulator forward kinematics in pose i, and the transformation from the sensor to the robot base frame ${}^{B}_{L}\boldsymbol{T}$. Hence, ${}^{O}_{B}\boldsymbol{T}$ is known and the unknown transformation matrix is ${}^{B}_{L}\boldsymbol{T} \in \mathbb{R}^{4\times4}$ that describes the location of sensor coordinate **{L}** to robot base **{B}**. Notably, ${}^{O}_{B}\boldsymbol{T}$ and ${}^{B}_{L}\boldsymbol{T}$ can be expressed by ${}^{O}_{B}R$ and ${}^{B}_{L}R$ matrices, representing rotations in $\mathbb{R}^{3\times3}$, and ^{B}t and ^{L}t, denoting translation vectors in $\mathbb{R}^{3\times1}$, respectively.

4. Methodology

The central idea is first to solve transformation matrix ${}^{O}_{L}\boldsymbol{T}$ by forming a linear system of equations. Considering a set of fixed scanners, we utilize a build plate on the robot's end effector for hand–eye calibration. Initially, the robot positions the build plate in the first location within the field of view of all line profilers. Subsequently, it executes sequential translation and rotation two times. This way, we gather six distinct measurements for the closed-form computation of 6 unknown parameters. The proposed data acquisition method for calibration encompasses the following steps:

1. Attach a plane to the robot end effector or utilize an existing planar artifact, such as an AM substrate.
2. Robot moves the plane to a pre-defined orientation R_i within the scanners' FoV and acquires measurements.
3. Execute a pure constant translation, T_i, along the plane's normal $\hat{n}_i$ and acquire measurements.
4. Repeat steps 2 and 3 two more times for the next orientations $i = 2$ and $i = 3$.
5. Utilize the proposed novel closed-form solution to estimate rotation and translation parameters.

The proposed calibration formulation is described in the following way, and it solves the unknown transformation in two steps: rotation and translation calibrations. The first step utilized measurements at three non-coplanar poses to determine the rotation uniquely. The orientation of the laser frame with respect to the base frame is established by the formulation in Section 4.1. With a known orientation, the sensor position with respect to the base frame can be calculated by employing measurements at three distinct orientations of the plane around a common pivot point (Section 4.2).

Fig. 3 represents the projection of laser rays on a plate artifact. A series of laser rays are projected over the laser opening angle θ, and the sensor matrix captures the reflection of laser rays. The raw data in the sensor coordinate frame is given as the position of each point, and its distance along x and y axes as $P_{2D} = \{(-x_n, z_{-xn}), \ldots, (x_n, z_{xn})\}$. The captured points from a build plate with roughness contain some noise and outliers. Hence, A 2D line is fitted on measured points to minimize the effect of measurement noise and build the plate's roughness. A ray is created from sensor origin p_0 at $0°$, called the central ray with direction $\hat{r}_c$, and the edge ray at an angle $\frac{\theta}{2}°$ is represented by $\hat{r}_e$.

The build plate can be represented as a plane, and the equation of a plane can be formed with the unit surface normal ($\hat{n}$), the position

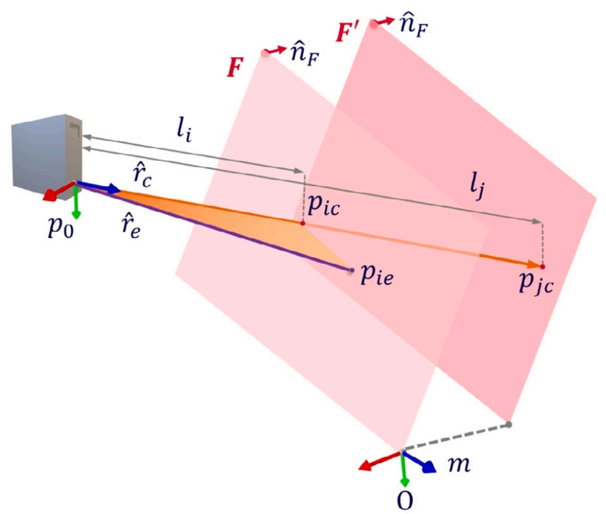


Fig. 4. Rotation calibration - Laser ray beam intersection with the plate before and after pure translation motion.

vector ($\vec{p}$), and the perpendicular distance d from the origin to the point in the plane as $\hat{n}\vec{p} = d$. The unit normal ($\hat{n}$) of the plane is known with the known robot pose ${}^{O}_{B}\boldsymbol{T}$.

4.1. Rotation calibration

Rotation calibration involves three plate profiles acquired before and after the robot moves the planar substrate along the plane's normal direction. This process requires reorienting the planar plate before each translational movement. It is crucial to ensure an adequate variance in the rotation angle to establish a full-rank system of equations. This would allow the unknown orientation of the sensor frame to be solved with respect to the global coordinate frame.

4.1.1. Solving the Z component ($\hat{L_Z}$)

Fig. 4 shows the calibration plane with a known orientation F, in which the plane is subject to translational movement, m, along the unit normal direction of plane $\hat{n}_F$. The vector formed between intersection points of the laser ray and plane at i and j positions, $\vec{P}_{ij} = (l_j - l_i)\hat{r_c}$, is projected onto the known normal vector of the plane $\hat{n}_F$ that relates to the magnitude of translation m as indicated by,

$$\hat{n_F} \cdot (l_j - l_i)\ \hat{r}_c = m \tag{2}$$

$$\begin{bmatrix} \hat{n}_{Fx} & \hat{n}_{Fy} & \hat{n}_{Fz} \end{bmatrix} \begin{bmatrix} \hat{r}_{cx} \\ \hat{r}_{cy} \\ \hat{r}_{cz} \end{bmatrix} = \frac{m}{l_j - l_i} = f \tag{3}$$

Repetition of this process for three different plane orientations, F, G, and H, produces a linear system of equations. This system is normalized and solved to obtain the unit ray direction $\hat{r}_c$ as described in Eq. (4).

$$\begin{bmatrix} \hat{n}_{Fx} & \hat{n}_{Fy} & \hat{n}_{Fz} \\ \hat{n}_{Gx} & \hat{n}_{Gy} & \hat{n}_{Gz} \\ \hat{n}_{Hx} & \hat{n}_{Hy} & \hat{n}_{Hz} \end{bmatrix} \begin{bmatrix} \hat{L_{Zx}} \\ \hat{L_{Zy}} \\ \hat{L_{Zz}} \end{bmatrix} = \begin{bmatrix} f \\ g \\ h \end{bmatrix} \tag{4}$$

where $\hat{r}_c$ represents $\hat{L}_Z$ which is Z axis of L coordinate frame with respect to workobject coordinate frame.

4.1.2. Solving Y and X components ($\hat{L}_Y$ and $\hat{L}_X$)

The edge ray $\hat{r}_e$ is calculated by an approach similar to that used for the center ray $\hat{r}_c$ in the previous section. Since both rays lie on the same laser plane, the Y component, $\hat{L}_Y$, is simply derived via the cross product of $\hat{r}_c$ and $\hat{r}_e$ and then it is normalized:

$$\hat{L}_Y = \hat{r}_c \times \hat{r}_e \ / \ |\hat{r}_c \times \hat{r}_e| \tag{5}$$

Finally, $\hat{L}_X = \hat{L}_Y \times \hat{L}_Z$ component is calculated.

4.1.3. Computing the rotation matrix

Employing Rodrigues's rotation formula, the rotations between corresponding axes of the X, Y, and Z-axes in the work object and sensor coordinate systems are computed. The compound rotation matrix is determined with Eq. (6):

$${}^{O}_{L}R = R_{(z-axis)} R_{(y-axis)} R_{(x-axis)} \tag{6}$$

4.2. Translational calibration

Fig. 5 illustrates two sets of rigid body transformations established through the laser ray intersection with planes F and H and planes F and G. Notably, the plane orientations converge at a common pivot point O, with known plane normals $\hat{n}_F$, $\hat{n}_G$, and $\hat{n}_H$ determined by the robot end-effector pose. The laser central rays intersect the plate at unknown points denoted as p_F, p_G, and p_H. The distances between point p_F and p_H and between point p_F and p_G are represented as Δl_{FH} and Δl_{FG}, respectively, and are established through scanner measurements.

The objective is to ascertain the translational component of the homogeneous transformation, representing the position of the origin of the laser frame relative to the object frame. The rigid body transformation of ${}^{O}\vec{p}_0$ is expressed through two distinct paths between frames O and L, illustrated in Figs. 5(a) and 5(b) as follows:

$${}^{O}\vec{p}_0 = t\widehat{FG} + \vec{u_F} - l\hat{r}_c \ , \tag{7}$$

$${}^{O}\vec{p}_0 = s\widehat{FH} + \vec{v_F} - l\hat{r}_c \ , \tag{8}$$

$$\widehat{FG} = \hat{n}_F \times \hat{n}_G, \widehat{FH} = \hat{n}_F \times \hat{n}_H \ , \tag{9}$$

where t and s represent unknown scalar parameters. The vectors $\vec{u_f}$ and $\vec{v_f}$ are oriented perpendicular to the extensions of $\widehat{FG}$ and $\widehat{FH}$ at points $\vec{p_u}$ and $\vec{p_f}$, respectively. Both vectors extend towards the point p_F. Although the directions of $\vec{u_f}$ and $\vec{v_f}$ are known, their magnitudes remain undetermined. The ensuing geometric solution outlines the proposed method for ascertaining the magnitudes of $\vec{u_f}$ and $\vec{v_f}$, and subsequently, the values of t and s.

In relation to $\vec{u_f}$, the first geometric relationship is depicted in Fig. 5(a). The intersection point p_G and the associated vectors are projected onto an imaginary plane I, which traverses through p_u, with a normal n_I parallel to $\widehat{FG}$. Consequently, $\vec{u_f}$ forms a triangle with the projected vectors $\vec{\Delta l}'_{FG}$ and $\vec{u}'_G$. The use of primes (') and dashed lines denotes the projected components of the original vectors, as shown in Fig. 5(a). Subsequently, the magnitude and direction of $\vec{u_f}$ are computed employing Eq. (10) by applying the law of sines to the projected triangle in the plane.

$$\begin{aligned} |\vec{u}_F| &= |\vec{\Delta l}'_{FG}| \frac{sin\ (\beta)}{sin\ (\alpha)} = |\vec{\Delta l}'_{FG}| \frac{|\widehat{\Delta l'}_{FG} \times \hat{u}'_G|}{|\hat{u}_F \times \hat{u}'_G|} \ , \\ \hat{u}_F &= \hat{n}_I \times \hat{n}_F \ , \\ \hat{u}'_G &= \hat{n}_I \times \hat{n}_G \ , \\ \vec{\Delta l}'_{FG} &= \vec{\Delta l}_{FG} - (\vec{\Delta l}_{FG} \cdot \hat{n}_I)\ \hat{n}_I \end{aligned} \tag{10}$$

Likewise, the determination of $\vec{v}_F$ follows a comparable process, Fig. 5(a), encompassing planes F and H:

$$\begin{aligned} |\vec{v}_F| &= |\vec{\Delta l}'_{FH}| \frac{sin\ (\delta)}{sin\ (\gamma)} = |\vec{\Delta l}'_{FH}| \frac{|\widehat{\Delta l'}_{FH} \times \hat{v}'_H|}{|\hat{v}_F \times \hat{v}'_H|}, \\ \hat{v}_F &= \hat{n}_{II} \times \hat{n}_F \ , \\ \hat{v}'_H &= \hat{n}_{II} \times \hat{n}_H \ , \\ \vec{\Delta l}'_{FH} &= \vec{\Delta l}_{FH} - (\vec{\Delta l}_{FH} \cdot \hat{n}_{II})\ \hat{n}_{II} \end{aligned} \tag{11}$$

With known vectors of $\vec{u}_F$ and $\vec{v}_F$, expressing Eq. (12) in matrix form decomposing each component, we get Eq. (13). Eq. (13) is an inconsistent matrix equation, and the best-approximated value of t and s is determined with the least-square solution. This process leads to the determination of the unknown vector ${}^{O}\vec{p_F}$.

$$t\widehat{FG} + \vec{u}_F = s\widehat{FH} + \vec{v}_F \tag{12}$$

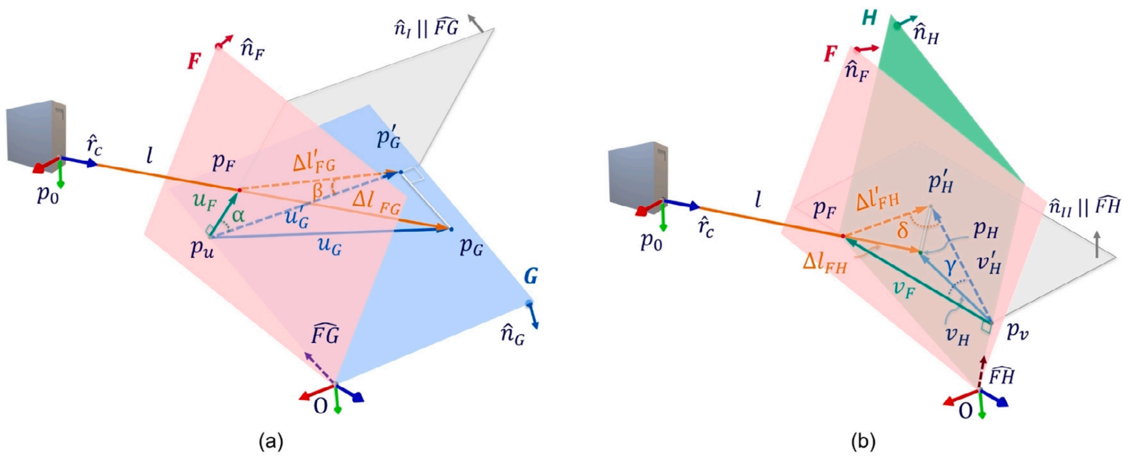


Fig. 5. Translation calibration: illustration of two sets of rigid body transformations through laser rays intersection with (a) planes F and H and (b) planes F and G. Planes I and II are imaginary planes employed for solving $\hat{u}_f$ and $\hat{v}_f$.

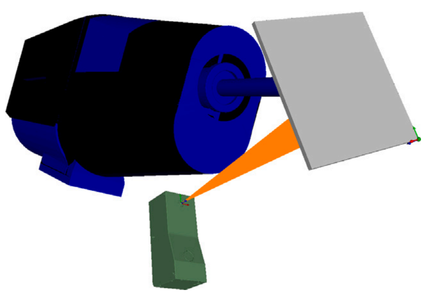


Fig. 6. A simulation environment: created to collect the virtual laser scanner data, test the method, and simulate the effect of individual error sources.

$$\begin{bmatrix} \widehat{FG}_x & -\widehat{FH}_x \\ \widehat{FG}_y & -\widehat{FH}_y \\ \widehat{FG}_z & -\widehat{FH}_z \end{bmatrix} \begin{bmatrix} t \\ s \end{bmatrix} = \begin{bmatrix} (\vec{v}_{Fx} - \vec{u}_{Fx}) \\ (\vec{v}_{Fy} - \vec{u}_{Fy}) \\ (\vec{v}_{Fz} - \vec{u}_{Fz}) \end{bmatrix} \tag{13}$$

5. Simulation and calibration accuracy

A custom simulator was developed to test the proposed method and evaluate the impact of plate roughness, plate flatness, TCP calibration error, and robot pose deviations on calibration accuracy. This simulation environment was created using C++ with OpenGL and the Eigen library. It supports importing and simulating robots, sensors and 3D plates, as illustrated in Fig. 6. A key feature of our simulation environment is its ability to acquire laser profiler data through ray casting. The scanner emitted rays onto the plate, returning profile points akin to an actual laser profiler. The algorithm then processed these simulated profiles to calculate the transformation matrix. This capability enabled us to execute automated toolpaths and collect virtual point cloud data. The collected point cloud data served as input for our method, which subsequently determined the sensor pose. As such, the simulation offered a controlled environment to isolate and study the effect of deviation in robot movement.

Theoretical accuracy: The closed-form solution in simulation yields exact results using noise-free data, a perfect plane, and an error-free robot pose. Despite this ideal scenario, round-off errors can occur during mathematical operations. We calculated the deviation of the closed-form solution's calibration results from the known calibration pose to quantify the theoretical error, demonstrating a negligible error within the nanometer range.

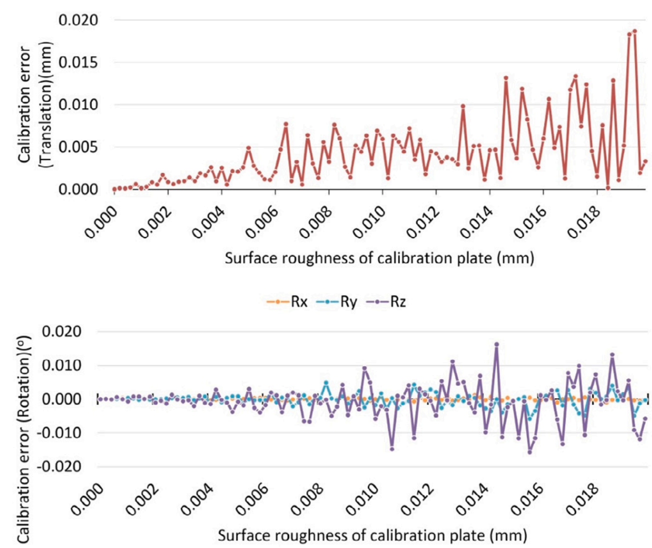


Fig. 7. The translation and rotation error in calibration result due to the surface roughness of the calibration plate.

Calibration accuracy due to plate roughness/flatness: The analysis was extended to examine the impact of surface roughness and flatness on the calibration results. A combined sinusoidal wave was introduced to the distance (z) along the length of the profile, described by the equation $A \sin\left(\frac{2\pi x}{\lambda} + \phi\right)$, where x represents the profile position, λ is the wavelength, A is the amplitude, and ϕ is the phase. The effects of the plate's surface roughness and flatness errors on the calibration results are illustrated in Figs. 7 and 8.

Calibration accuracy due to TCP calibration Error and robot pose deviations: we explored the effects of robot pose deviation and TCP calibration error. During our simulations, we defined a robot toolpath to control the movement of the plate's STL model. Concurrently, errors were intentionally introduced to the robot's pose during data acquisition. Calibration was then performed using simulated data, with calibration error defined as the discrepancy between the known and calculated poses. We observed that deviations in the robot's pose resulted in incorrect rotational calculations, which subsequently impacted the translational transformations. Fig. 9 illustrates the impact of robot pose deviation on calibration error. Furthermore, TCP deviation was induced with a specified magnitude and random unit normal direction across various orientations. A systematic increase in TCP deviation

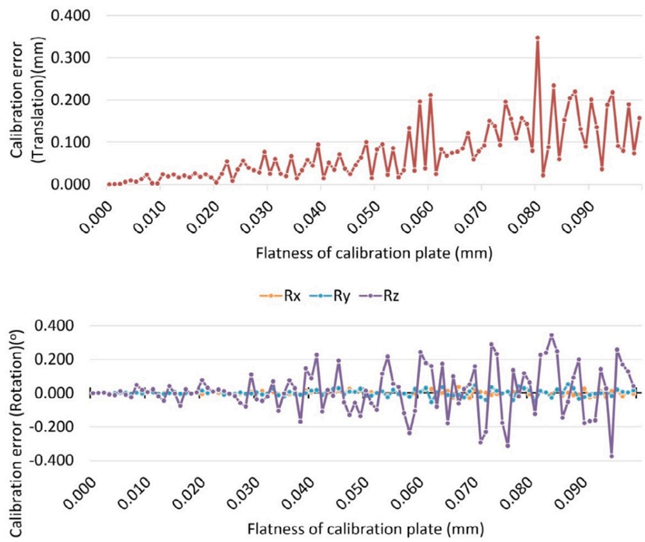


Fig. 8. The translation and rotation error in calibration result due to the flatness error of calibration plate.

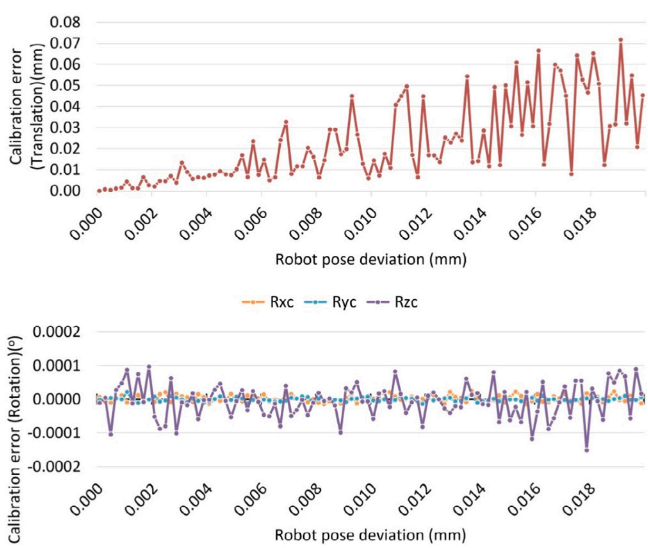


Fig. 9. The translation and rotation error in calibration result due to the robot pose deviation.

was applied across all poses to simulate the calibration error resulting from TCP calibration inaccuracies. The effect of TCP calibration error on hand–eye calibration error is demonstrated in Fig. 10.

Results demonstrated the effectiveness of the calibration method in the case of small roughness of the plate, inaccuracy of the flatness, and errors in robot pose. However, for a precise calibration, it is essential to use a calibrated plate and ensure the accuracy of the TCP calibration, as those can impact the calibration accuracy.

6. Experimental setup

The proposed calibration methodology was implemented and experimentally tested in a robotic cell equipped with an ABB 6-DoF industrial robot (IRB4600). The robot has a pose accuracy and repeatability of 0.02 mm and 0.06 mm, respectively. The sensory system comprises three fixed laser line scanners located at different viewpoints: one Micro-Epsilon 2700-100 with a z-resolution of 15 μm and a standard range 350–450 mm and two Micro-Epsilon 2900-100 with a z-resolution of 12 μm and a standard range 190–290 mm. All the

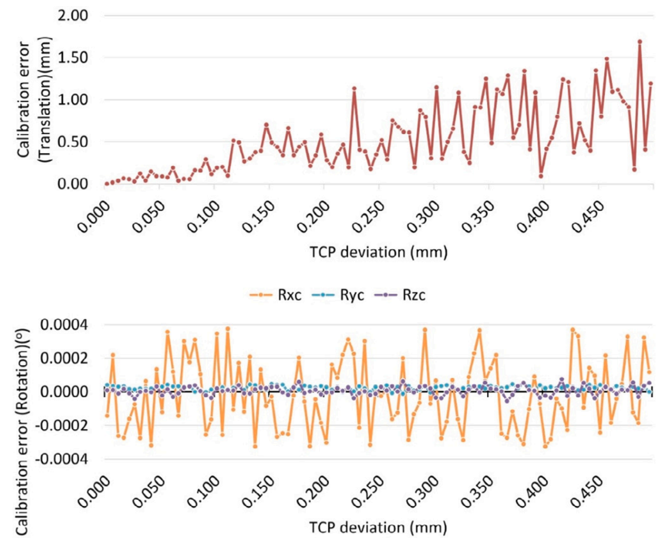


Fig. 10. The translation and rotation error in calibration result due to the TCP calibration error.

scanners focused on a common region forming a triangular shape that can facilitate their use as a robotic scanner or in-process 3D scanning in robotic additive manufacturing. This configuration allowed the plate to always be within the visual scope of the scanner. A reference resolution on the x-axis was chosen as 640 points/profile. The calibration artifact chosen for this experiment was a 200 mm×200 × 6 mm rectangular flat plate. In a robotic cell, a similar plate can be used as the build plate to manufacture 3D objects, such as in cold spray additive manufacturing [36]. The calibration plate was attached to a flange connected to the end effector. In this setup, the cold spray gun is used as a stationary tool, and TCP is defined as a point 30 mm distance from the end of the nozzle.

7. Calibration process

The proposed hand–eye calibration method employed a rectangular build plate at six distinct poses, incorporating consecutive rotational and translational displacements as illustrated in Fig. 11. Two sets of calibration experiments and validations were performed and discussed in the results section, one for a single scanner and the other involving three scanners triggered simultaneously. For each experiment, calibration was performed using the same plane poses.

Within the setup depicted in Fig. 13, the plate's initial pose was set to $T_0 = (100, 100, 0)$, from the bottom left corner of the plate, ensuring visibility to all three scanners. The center of the build plate is the common pivot point for different orientations. Subsequently, the plate underwent displacements by (R_1, T_0), (R_1, T_1), (R_2, T_0), (R_2, T_1), (R_3, T_0), and (R_3, T_1). The rotational angles were defined as $R_1 = (25°, 0, 0)$, $R_2 = (-25°, 0, -25°)$, and $R_3 = (-25°, 0, 25°)$, while $T_1 = T_2 = T_3 = 10$ mm along the plate's normal at R_1, R_2, and R_3. The selected poses ensured variation within the constraints of the setup, sensors's FoV, and robot reachability, and avoiding collisions between the plate, tool, and robot. We utilized in-house simulation software for pose selection and ABB RobotStudio® for path planning.

Scanners were automatically triggered in a cascade fashion after moving the plate to each location. An in-house trigger box created a cascade signal for each scanner, and a delay was introduced between measurements to mitigate potential interference from the previous scanner's laser reflection. Each 2D line profile measurement consisted of densely sampled points from laser ray intersecting the plane. However, sparse outliers must be removed, especially on edges, with a radius outlier remover (ROR). Then, a 2D line was fitted to these

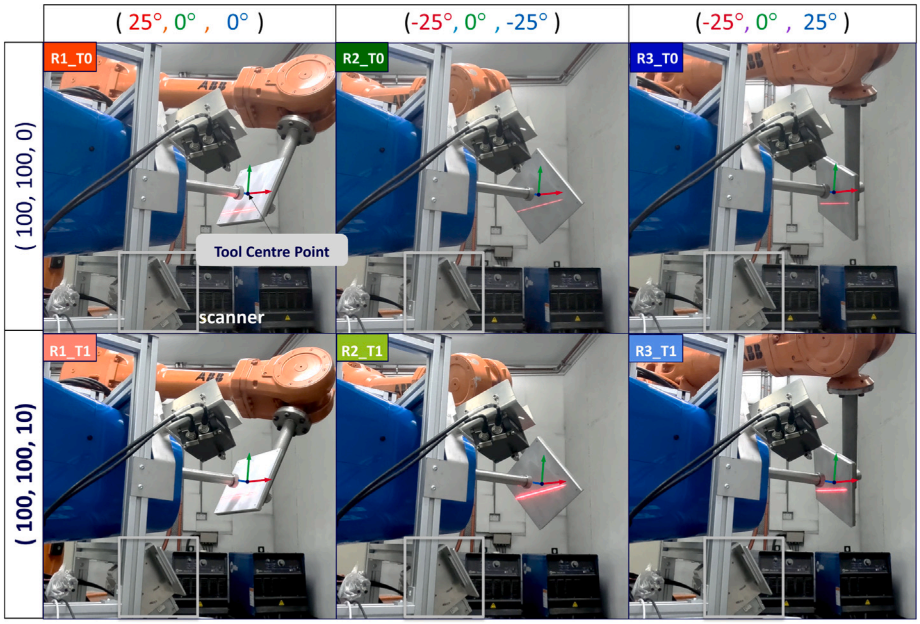


Fig. 11. Calibration data collection: acquiring profiles of the plate at six distinct poses. $R_i\_T_j$ indicates the plate movement concerning ith rotation around pivot point and jth translational displacement. $R_1 = (25^o, 0, 0)$, $R_2 = (-25^o, 0, -25^o)$, $R_2 = (-25^o, 0, 25^o)$, $T_1 = T_2 = T_3 == (100, 100, 10)$ mm along the plate normal at R_1, R_2 and R_3. Plate initial pose is $P_0 = (100, 100, 0)$.

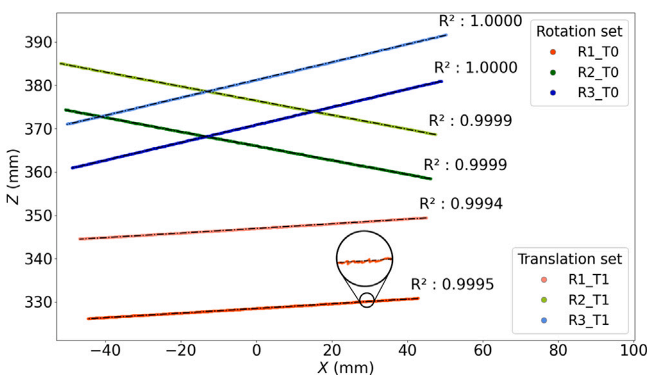


Fig. 12. Line fitted on acquired profiles for calibration.

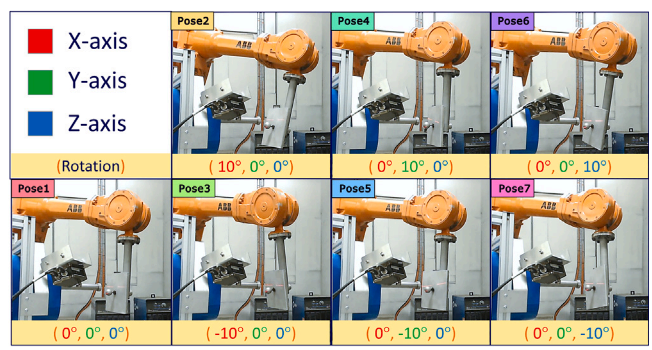


Fig. 13. Validation experiments via scanning a moving calibrated sphere of 34.921 mm diameter.

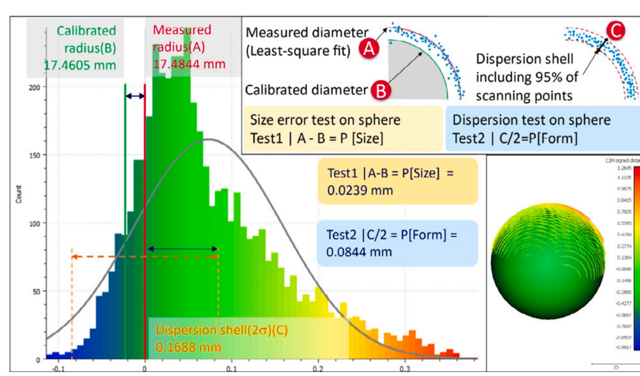


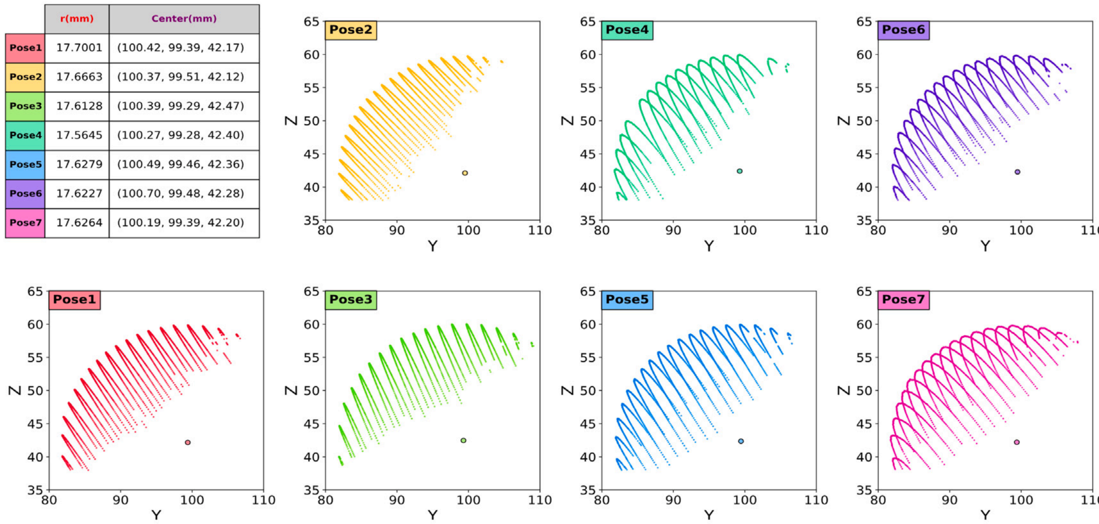
Fig. 14. The distribution of points from scanning the precision sphere of 17.4605 mm. Test1 represents the size error test on the sphere, and test2 is the dispersion test on the sphere (form error).

sampled points, and the R-squared results of the line fittings are shown in Fig. 12. The R-squared error is close to 1, which indicates accurate line fitting. Additionally, the actual robot pose for each measurement was recorded. Given the above measurement, the calibration problem was solved with the proposed formulations in Section 4.

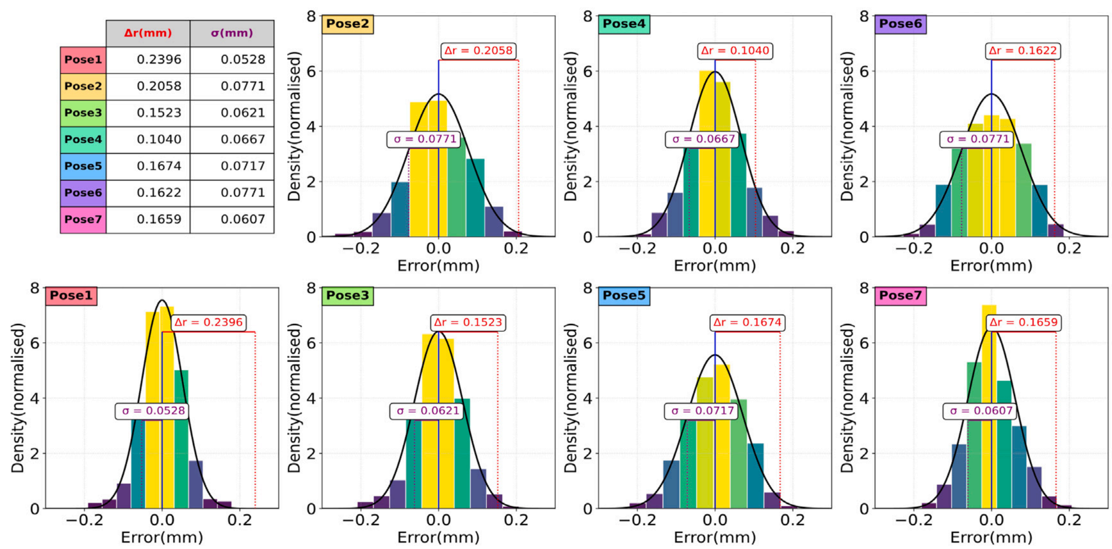
8. Validation via sphere reconstruction

A precision sphere with a 17.4605 mm radius was bolted on the center of the plate and moved vertically upward by the robot while line profile measurements were recorded. This was repeated for seven different plate orientations, as shown in Fig. 13. Utilizing the estimated transformation parameters, the process of 3D reconstruction is principally addressed through Eq. (1). The reconstruction of points on the sphere served to validate the outcome and evaluate the method's appropriateness.

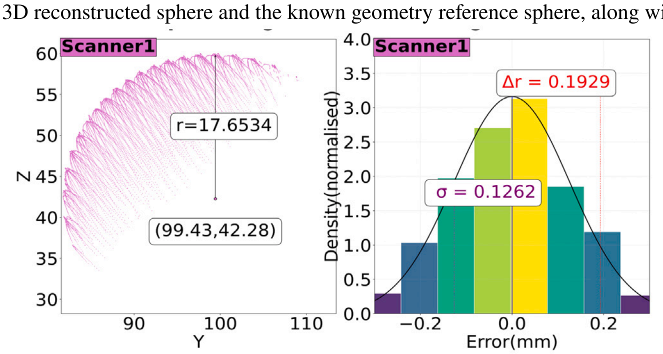
The choice to use a sphere was based on the following reasons: (a) Sphere validation is accepted by metrology standards (ISO 10360-8). (b) A Sphere's Lambertian surface avoids inaccuracies in measurement from reflection (c) The spherical shape allows the scanner to be tested from multiple angles and orientations to test positional accuracy and radial errors. Fig. 14 represents the size error and dispersion test on the sphere. The sphere size error was 0.0239 mm, and the form error was measured to be 0.0844 mm, as reconstructed by the calculated transformation matrix.



(a) Reconstructed 3D points and calculated sphere center and radius from sphere regression



(b) Error between the 3D reconstructed sphere and the known geometry reference sphere, along with standard deviations



(c) single scanner sphere regression result utilizing all data

Fig. 15. Calibration validation involving single sensor: a sphere was fitted based on seven groups of scanning a precision ball. Top: estimated radius and center for each set, Middle: mean error and std, Bottom: results for combined data.

A 3D point cloud was reconstructed for each validation dataset by registering 2D line profiles, followed by a least squares sphere regression to the points. The fitted sphere's calculated average radius and center are detailed in the table within Fig. 15(a). Fig. 15(b) illustrates the error between the 3D reconstructed sphere's radius and the known geometry reference sphere's radius, along with standard deviations. The radius deviation spans from 0.1040 mm to 0.2396 mm.

In the subsequent series of experiments, the calibration performance of three scanners was evaluated. A cascade-triggering method was employed to prevent interference among the scanners. Sphere validation followed a procedure akin to the preceding experiment. The seven sets of reconstructed point clouds were amalgamated, and each scanner performed sphere fitting independently. Fig. 16(a) illustrates the results of sphere reconstruction for each sensor, featuring the fitted center,

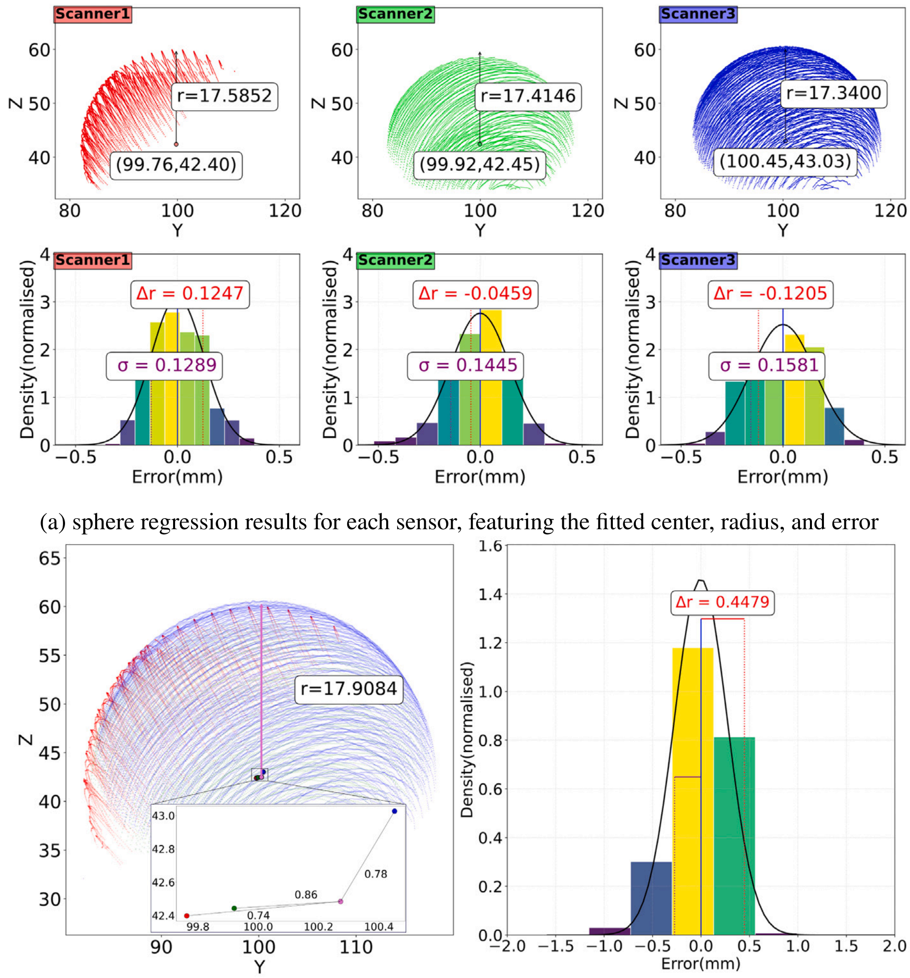


Fig. 16. Calibration validation involving three sensors: Top: results for each sensor Bottom: results for combined data.

Table 2
The first two columns represent the radius and standard deviation of the fitted sphere for three laser profilers. The last column is the difference between the fitted radius and the standard radius(17.4605 mm).

Sensor	radius(mm)	std.(σ(mm))	error (Δr)(mm)
1	17.5852	0.1289	0.1247
2	17.4146	0.1445	−0.0459
3	17.3400	0.1581	−0.1205

radius, and its error. Table 2 additionally provides insights into the error between the fitted centers of individual scanners and the center of the combined sphere. Fig. 16(b) also indicates similar results in the case of combining data from all three sensors.

9. Discussion

The sphere scanning experiments and regression fitting have successfully demonstrated the effective and accurate calibration of both single and multiple 2D profilers, leading to precise 3D reconstruction. In the case of a single sensor, the results showed an average radius deviation of 0.1929 mm. This error ranged from 0.1040 mm to 0.2396 mm, depending on the initial angle of the shape in relation to the sensor. It is important to note that when the sphere is in an extreme relative orientation to the sensor, some laser reflections may not be received by the sensor, resulting in reduced measurement accuracy. The scanner achieves its highest accuracy when the sensor is perpendicular to the measured surface. Similarly, the average radius deviation of the regression model utilizing three sensors is indicated as 0.1247 mm, −0.0459mm, and −0.1205mm.

The result from the single plane closed form calibration method is promising as it does not require any given initial parameters. However, a direct comparison of this method is hard to achieve with the current configuration, where scanners are not perpendicular to the measured surface. The manufacturer's instruction manual suggests the scanner be perpendicular to the measuring surface within the standard measuring range to achieve the mentioned scanner accuracy. Also, the robot's pose variability is constrained in this setup. Selecting a robot pose with enough variation is crucial for the solution to be convergent and correct. This method expands the possibility of calibrating multiple sensors simultaneously with the same robot pose.

The accuracy of the calibration method is affected by the tool center point (TCP) calibration accuracy, robot positional accuracy, scanner accuracy, and manufacturing tolerance of the validation sphere. Decoupling the influences of each component on hand–eye calibration results presents a significant challenge. However, our simulation study successfully isolated each error source, demonstrating their individual impacts on the calibration results.

Prior to hand–eye calibration, we performed TCP frame calibration, which is highly dependent on the operator's expertise. Inaccuracies in the TCP calibration could directly contribute to the overall hand–eye calibration error. We have performed TCP calibration before hand–eye calibration.

Pose accuracy in industrial robots varies based on factors such as robot model and payload, typically ranging between 10–50 μm. The hand–eye calibration process is particularly sensitive to rotational and translational errors during the robot's manipulation of the calibration plate. To reduce the impact of pose deviations, we utilized the executed pose as input to the proposed closed-form calibration method rather than the expected/programmed pose.

To maintain smooth robot motion, one may introduce tolerances in toolpath, which we avoided by utilizing no zone data and opting for joint motion over linear motion.

Another error source is the quality of the calibration plate. Although the mathematical foundation assumes an ideal plane, practical calibration plates have inherent tolerances. Surface roughness and flatness affect the point cloud acquired by the laser profiler. Our method addresses these deviations by employing a two-step approach: (a) resampling the acquired profile points using an average filter of size 5, and (b) fitting a 2D line to the surface points.

While the laser scanner offers accuracy within a standard measuring range of a few tens of micrometers, environmental factors such as surface reflectivity and material properties introduce additional errors. We mitigated these by adjusting exposure settings and applying anti-reflective chalk to the plate before data acquisition.

The plane's orientation affects the surface normal used in calculations. In our setup, the calibration plate is rigidly attached to the robot flange, ensuring stability and accuracy. We calibrated the tool frame using the edge of the plate as reference points, precisely aligning it with the work-object frame.

As per the validation procedure, the precision sphere, with diameter tolerance and sphericity within the nanometer range and calibration errors, resulted in a measured sphere form error of 0.0844 mm that was reconstructed using the calculated transformation matrix.

Although an explicit error propagation study was not conducted in this work, such an analysis would be a valuable addition to future research. By systematically evaluating and modeling the effects of each error source, including kinematic modeling, future studies could identify high-sensitivity areas and enable more precise calibration adjustments, thereby enhancing and optimizing the accuracy of the hand–eye calibration approach.

10. Conclusion

This paper presents a new technique for calibrating a 2D laser profile scanner. The calibration object used in this study was a deposition plate attached to an industrial robot. By applying concepts from 3D projective geometry, a closed-form solution was developed. To evaluate the proposed method, a simulation environment was created with a virtual scanner and calibration plane. The method's accuracy was assessed through simulations and actual data obtained from the scanner. The calibration process is fast, allowing simultaneous calibration of multiple sensors without requiring additional poses. Additionally, the process does not require expert knowledge and can be performed by anyone applying this automated calibration process. This approach is particularly suitable for calibrating multiple scanners for in-process monitoring in cold spray additive manufacturing (CSAM). The achieved accuracy is sufficient for this type of additive manufacturing, which requires millimeter-level precision. In future work, an optimization-based method will be incorporated to further refine the accuracy achieved with this technique.

CRediT authorship contribution statement

Subash Gautam: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Hans Lohr:** Writing – review & editing, Software, Methodology, Conceptualization. **Alejandro Vargas-Uscategui:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Conceptualization. **Peter C King:** Writing – review & editing, Resources, Methodology, Conceptualization. **Alireza Bab-Hadiashar:** Writing – review & editing, Methodology, Conceptualization. **Ivan Cole:** Writing – review & editing, Methodology, Conceptualization. **Ehsan Asadi:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Video: https://doi.org/10.25919/byjw-v161

Data availability

The authors do not have permission to share data.

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